Project Idea 3: Self-Healing Cloud Deployment Pipelines

Description

Create an agentic framework for managing cloud deployment pipelines that integrates proactive diagnosis and automated repair. This project builds on self-healing DevOps concepts, specifically targeting deployment failures, configuration drifts, and runtime performance issues.

Core Objectives

- Develop a pipeline integration that triggers diagnostic agents upon failure.
- Implement a 'chaos' injection agent to stress-test system resilience.
- Create a repair agent capable of rolling back or patching configurations autonomously.

Final Outcome

A highly resilient CI/CD framework that autonomously detects deployment failures, diagnoses the root cause, and applies corrective actions without human intervention.